

Appendix

Time-to-Event Estimation with Unreliably Reported Events in Medicare Health Plan Payment

1. Considerations for underreporting estimand

The underreporting estimand is used to uniformly shift cumulative incidence in the comparison group ($g = 0$, e.g. Traditional Medicare, TM) only. Aligned with prior work^{1,2}, we recommend choosing reference events for underreporting estimation that are expected to persist over time (e.g. paraplegia, human immunodeficiency virus/acquired immunodeficiency syndrome). However, because reference events should be distinct from Hierarchical Condition Categories (HCCs) used for the other forms of estimation we propose the exact set of reference events used by researchers will vary. It is also possible that some chronic HCCs are not coded in sequential monitoring periods for legitimate reasons (e.g. if a beneficiary does not receive care for the condition in both periods).

In addition, we note that prior work has found that underreporting is consistently present in TM, but the degree varies by HCC and by year^{1,2}. This also means that underreporting estimates will likely vary depending on researchers' choice of reference events. However, current coding intensity estimation approaches overwhelmingly disregard underreporting overall, so our proposed estimator is a helpful first step towards better accounting for this issue and avoiding overestimation of coding intensity between Medicare programs. Future work in undercoding estimation could better account for heterogeneity in coding behaviors.

2. Variance and covariance of estimators

Variance and (if applicable) covariance are reported by estimator below.

2.1. $\hat{F}_s(t)$

$\hat{F}_s(t)$ is the estimator for event-specific cumulative incidence. The variance for $\hat{F}_s(t)$ is $\widehat{\text{var}}(\hat{F}_s(t)) = \sum_{t_l \leq t} \widehat{\text{var}}(\hat{\theta}(t_l)) + 2 \sum_{t_l < t} \sum_{t_l < t_i \leq t} \widehat{\text{cov}}(\hat{\theta}(t_l), \hat{\theta}(t_i))$, and covariance is $\widehat{\text{cov}}(\hat{F}_s(t), \hat{F}_s(u)) = \widehat{\text{var}}(\hat{F}_s(t)) + \sum_{t_l \leq t} \sum_{t < t_i \leq u} \widehat{\text{cov}}(\hat{\theta}(t_l), \hat{\theta}(t_i))$. Here t_l indicates an event reporting time prior to t_i and u is an arbitrary time distinct from t . $\hat{\theta}(t_i)$ has variance $\widehat{\text{var}}(\hat{\theta}(t_i)) = (\hat{\theta}(t_i))^2 \times ((r_i - d_{si})/(d_{si}r_i) + \sum_{t_l < t_i} (d_l/(r_l(r_l - d_l))))$ and covariance $\widehat{\text{cov}}(\hat{\theta}(t_i), \hat{\theta}(t_l)) = \hat{\theta}(t_i)\hat{\theta}(t_l)(-(1/r_i) + \sum_{t_l < t_i} (d_l/(r_l(r_l - d_l))))^3$, where $d_l = \sum d_{s,l}$ is the reported count of events of any subtype at t_l , $d_{s,l}$ is analogous to $d_{s,i}$, and r_l is the count of beneficiaries at risk at t_l .

2.2. $\hat{\mu}_s(\tau)$

$\hat{\mu}_s(\tau)$ is the estimator for $\psi = \mu_{s,g=1}(\tau) - \mu_{s,g=0}(\tau)$, or the restricted mean time lost for event s up in a single monitoring period with end time τ . This estimator has variance $\widehat{\text{var}}(\hat{\mu}_s(\tau)) = \sum_{t_i < \tau} (t_{i+1} - t_i)^2 \widehat{\text{var}}(\hat{F}_s(t_i)) + 2 \sum_{t_i < \tau} \sum_{t_l < t_i} (t_{i+1} - t_i)(t_{l+1} - t_l) \widehat{\text{cov}}(\hat{F}_s(t_i), \hat{F}_s(t_l))$.

2.3. $\hat{\psi}$ and $\hat{\psi}^*$

$\hat{\psi}$ is the estimand for $\psi = \mu_{s,g=1}(\tau) - \mu_{s,g=0}(\tau)$, or the difference in mean time without event across groups within one monitoring period. $\hat{\psi}^*$ is the estimate of the same estimand with underreporting correction. The variance of $\hat{\psi}$ is $\widehat{\text{var}}(\hat{\psi}) = \widehat{\text{var}}(\hat{\mu}_{s,g=1}(\tau)) + \widehat{\text{var}}(\hat{\mu}_{s,g=0}(\tau))$, as groups are assumed to be independent. It does not change for $\hat{\psi}^*$.

2.4. $\hat{\psi}_M$ and $\hat{\psi}_M^*$

$\hat{\psi}_M$ estimates ψ across sequential monitoring periods $\psi_M = \psi_m - \psi_{m-1}$, and $\hat{\psi}_M^*$ does the same with an underreporting correction in the comparison group. The variance of $\hat{\psi}_M$ is $\widehat{\text{var}}(\hat{\psi}_M) = \widehat{\text{var}}(\hat{\psi}_m) + \widehat{\text{var}}(\hat{\psi}_{m-1})$. Although the components of this estimator come from the same fixed population, the correlation between component estimates is assumed to be negligible because we are comparing differences in restricted mean time lost across independent groups estimated in disjoint time intervals. Therefore, covariance is zero. The variance remains unchanged for $\hat{\psi}_M^*$.

2.5. $\hat{\omega}$

$\hat{\omega}$ estimates possible severity-based upcoding in one monitoring period. The variance of $\hat{\omega}$ is $\widehat{\text{var}}(\hat{\omega}) = \widehat{\text{var}}(\hat{\omega}_{g=1}) + \widehat{\text{var}}(\hat{\omega}_{g=0})$, where $\widehat{\text{var}}(\hat{\omega}_g) = \widehat{\text{var}}(\hat{\mu}_{s=k}(\tau)) + \widehat{\text{var}}(\hat{\mu}_{s=1}(\tau)) - 2\widehat{\text{cov}}(\hat{\mu}_{s=k}(\tau), \hat{\mu}_{s=1}(\tau))$ and $\widehat{\text{cov}}(\hat{\mu}_{s=k}(\tau), \hat{\mu}_{s=1}(\tau)) = \sum_i \sum_j \widehat{\text{cov}}(\hat{F}_k(t_{i-1}), \hat{F}_1(t_{j-1}))$ for distinct event times t_i and t_j . Here, t_i corresponds to event reporting increments in the cumulative incidence function for severity level k , while t_j corresponds to the equivalent for severity level 1. Although event reporting times are the same, we write these using separate variables because covariance is estimated between all pairs of increments for these two severity levels.

3. Summary statistics from All of Us survey data

Table 1: **Included surveys from All of Us.** These data represent included survey respondents with at least one self-reported health condition that mapped to a Hierarchical Condition Category (HCC) in Version 28 of the Medicare Advantage risk adjustment algorithm. Only sets of co-occurring HCCs with more than 21 respondents (which also excludes several sets with more than 20 respondents) were used in any summary tables to comply with the All of Us Data and Statistics Dissemination Policy.

All of Us Survey Title	Number of Unique Respondents
Overall Health	156
Personal and Family Health History	15692

Table 2: **Number of respondents by HCC with available survey questions.** These data represent included survey respondents with at least one self-reported health condition that mapped to a Hierarchical Condition Category (HCC) in Version 28 of the Medicare Advantage risk adjustment algorithm. Only sets of co-occurring HCCs with more than 21 respondents (which also excludes several sets with more than 20 respondents) were used in any summary tables to comply with the All of Us Data and Statistics Dissemination Policy.

Version 28 HCC	Number of Respondents
HCC1	546
HCC20	339
HCC21	5775
HCC22	260
HCC23	6772
HCC35	156
HCC38	2114
HCC51	4283
HCC62	156
HCC64	100
HCC65	546
HCC77	156
HCC78	511
HCC93	1131
HCC109	1018
HCC155	1467
HCC182	47
HCC221	156
HCC226	73
HCC228	399
HCC238	1635
HCC249	373
HCC264	62
HCC267	123
HCC276	156
HCC280	273
HCC300	587
HCC327	31
HCC328	110
HCC398	808

Table 3: **Distribution of HCC-relevant conditions per person.** These data represent included survey respondents with at least one self-reported health condition that mapped to a Hierarchical Condition Category (HCC) in Version 28 of the Medicare Advantage risk adjustment algorithm. Only sets of co-occurring HCCs with more than 21 respondents (which also excludes several sets with more than 20 respondents) were used in any summary tables to comply with the All of Us Data and Statistics Dissemination Policy.

Number of HCC-Relevant Conditions Per Person	Number of Respondents
1	5636
2	6657
3	3163
4	236
5	156

Table 4: **Respondent age at survey response.** The following tables show respondent information using categories provided by All of Us. Not all respondents responded to all questions so counts may vary. The rows in each table correspond to available fields in All of Us. These characteristics are not displayed jointly because that would result in sample sizes too small to comply with the All of Us Data and Statistics Dissemination Policy.

Age Group (years)	Number of Respondents	Percent of Respondents (%)
[65-70)	5088	32
[70-75)	5767	36
[75-80)	3383	21
[80-85)	1362	9
[85-90)	250	2

Table 5: **Respondents' self-reported sex at birth.** The following tables show respondent information using categories provided by All of Us. Not all respondents responded to all questions so counts may vary. The rows in each table correspond to available fields in All of Us. These characteristics are not displayed jointly because that would result in sample sizes too small to comply with the All of Us Data and Statistics Dissemination Policy.

Self-Reported Sex at Birth	Number of Respondents	Percent of Respondents (%)
Female	8461	53
Male	6975	44
No matching concept	269	2
Not male, not female, prefer not to answer, or skipped	143	1

Table 6: **Respondents' self-reported ethnicity.** The following tables show respondent information using categories provided by All of Us. Not all respondents responded to all questions so counts may vary. The rows in each table correspond to available fields in All of Us. These characteristics are not displayed jointly because that would result in sample sizes too small to comply with the All of Us Data and Statistics Dissemination Policy.

Self-Reported Ethnicity	Number of Respondents	Percent of Respondents (%)
Hispanic or Latino	633	4
Not Hispanic or Latino	14442	91
Prefer not to answer	34	0
Skipped	639	4
Neither	100	1

Table 7: **Respondents' self-reported race.** The following tables show respondent information using categories provided by All of Us. Not all respondents responded to all questions so counts may vary. The rows in each table correspond to available fields in All of Us. These characteristics are not displayed jointly because that would result in sample sizes too small to comply with the All of Us Data and Statistics Dissemination Policy.

Self-Reported Race	Number of Respondents	Percent of Respondents (%)
Asian	260	2
Black or African American	752	5
White	13347	84
Another single population	49	0
More than one population	129	1
I prefer not to answer	34	0
Skipped	639	4
None indicated	538	3
None of these	100	1

4. Additional information about upcoding R package

This package is available under a MIT license, a copy of which is included in the package repository. A brief tutorial on specific package functions is available in the package’s Github repository.

4.1. Baseline data simulation

We simulated co-occurring hierarchical condition categories (HCCs) using data from the All of Us study⁴. This study was designed to enroll a million participants across the United States, focusing especially on groups historically underrepresented in clinical and biomedical research⁴ and included questions that overlapped with many HCCs in the current version of the Centers for Medicare and Medicaid Services’ (CMS) Medicare Advantage (MA) risk adjustment algorithm (i.e. CMS-HCC Version 28, or V28). V28 HCCs (listed in Appendix Table 8) were manually mapped to Systemized Nomenclature of Medicine-Clinical Terms (SNOMED-CT) concept identifiers for survey questions (using version 7 of All of Us data; see this project’s Github repository for mapping). Then, survey responses to all available SNOMED-CT concepts were queried from the All of Us database. Included surveys, available respondent sociodemographic characteristics, and coverage of V28 HCCs are described in Section 5.

For each surveyed person, self-reported co-occurring V28 HCCs were extracted. This was then summarized into a table of unique co-occurring HCC sets and a count of the number of All of Us survey respondents that reported having these diagnoses. In line with the All of Us Data and Statistics Dissemination Policy, only sets of co-occurring HCCs with more than 21 respondents were exported from All of Us’s platform (the Researcher Workbench) to be used in analyses. This both omits all sets of 20 or fewer respondents and also excludes several sets of co-occurring HCCs with more than 20 respondents. Users also have the option to use alternate sets of co-occurring V28 HCCs for baseline data sampling if they prefer.

4.2. Upcoding and loss to follow up simulation

The package enables users to simulate two types of upcoding realistic to MA: any-available or severity-based, although the latter can only occur if an HCC has competing events. This upcoding is randomly split across a user-specified number of time points. Even though there is inherent right censoring in upcoded data (because only a proportion of all simulated individuals will be upcoded or coded at all), the package provides users with the option to include additional right censoring. This is meant to be representative of loss to follow up or death, both of which are known issues in following a population aged 65 years and older over time⁵. Users specify the proportion of loss to follow up they want to include, and a randomly selected set of rows (e.g., individuals) are right censored over each time point. Once someone is lost to follow up, they cannot be coded for any HCC in subsequent time periods.

4.3. Undercoding simulation

The package also separately enables undercoding. Here, users specify a proportion of all coded diagnoses (from simulated baseline data) to randomly remove across the entire data set. This occurs on a dataset-wide level because undercoding is a systemic issue in Traditional Medicare^{1,2} and so we do not assume that it impacts specific V28 HCCs disproportionately.

5. Additional information about version 28 Hierarchical Condition Categories (HCCs)

Table 8: **Hierarchical Condition Categories (HCCs) included in the Medicare Advantage risk adjustment algorithm version 28 (V28).** Each of the HCCs in this table are described as written in the Centers for Medicare and Medicaid Services’ (CMS) 2024 Medicare Advantage Advance Notice⁶. Competing events, if applicable, are also noted from the 2024 CMS Risk Adjustment Model Software and ICD-10 Mappings, specifically the Midyear/Final Model Software, which is in the SAS language⁷. If a set of competing events is not N/A, then, if the HCC in that row is coded, none of the competing event HCCs can also be coded for billing purposes. The hierarchy described in the “Competing Events” column can also be found in the upcoding R package.

HCC	HCC Description	Competing Events
HCC1	HIV/AIDS	N/A
HCC2	Septicemia, sepsis, systemic inflammatory response syndrome/shock	N/A
HCC6	Opportunistic infections	N/A
HCC17	Cancer metastatic to lung, liver, brain, and other organs; acute myeloid leukemia except promyelocytic	HCC18, HCC19, HCC20, HCC21, HCC22, HCC23
HCC18	Cancer metastatic to bone, other and unspecified metastatic cancer; acute leukemia except myeloid	HCC19, HCC20, HCC21, HCC22, HCC23
HCC19	Myelodysplastic syndromes, multiple myeloma, and other cancers	HCC20, HCC21, HCC22, HCC23
HCC20	Lung and other severe cancers	HCC21, HCC22, HCC23
HCC21	Lymphoma and other cancers	HCC22, HCC23
HCC22	Bladder, colorectal, and other cancers	HCC23
HCC23	Prostate, breast, and other cancers and tumors	N/A
HCC35	Pancreas transplant status	HCC36, HCC37, HCC38
HCC36	Diabetes with severe acute complications	HCC37, HCC38
HCC37	Diabetes with chronic complications	HCC38
HCC38	Diabetes with glycemic, unspecified, or no complications	N/A
HCC48	Morbid obesity	N/A
HCC49	Specified lysosomal storage disorders	N/A

HCC	HCC Description	Competing Events
HCC50	Amyloidosis, porphyria, and other specified metabolic disorders	N/A
HCC51	Addison's and Cushing's diseases, acromegaly, and other specified endocrine disorders	N/A
HCC62	Liver transplant status/complications	HCC63, HCC64, HCC65, HCC68
HCC63	Chronic liver failure/end-stage liver disorders	HCC64, HCC65, HCC68, HCC202
HCC64	Cirrhosis of liver	HCC65, HCC68
HCC65	Chronic hepatitis	N/A
HCC68	Cholangitis and obstruction of bile duct without gallstones	N/A
HCC77	Intestine transplant status/complications	HCC78, HCC80, HCC81
HCC78	Intestinal obstruction/perforation	N/A
HCC79	Chronic pancreatitis	N/A
HCC80	Crohn's disease (regional enteritis)	HCC81
HCC81	Ulcerative colitis	N/A
HCC92	Bone/joint/muscle/severe soft tissue infections/necrosis	N/A
HCC93	Rheumatoid arthritis and other specified inflammatory rheumatic disorders	HCC94
HCC94	Systemic lupus erythematosus and other specified systemic connective tissue disorders	N/A
HCC107	Sickle cell anemia (Hb-SS) and thalassemia beta zero	HCC108
HCC108	Sickle cell disorders, except sickle cell anemia (Hb-SS) and thalassemia beta zero; beta thalassemia major	N/A
HCC109	Acquired hemolytic, aplastic, and sideroblastic anemias	N/A
HCC111	Hemophilia, male	HCC112

HCC	HCC Description	Competing Events
HCC112	Immune thrombocytopenia and specified coagulation defects and hemorrhagic conditions	N/A
HCC114	Common variable and combined immunodeficiencies	HCC115
HCC115	Specified immunodeficiencies and white blood cell disorders	N/A
HCC125	Dementia, severe	HCC126, HCC127
HCC126	Dementia, moderate	HCC127
HCC127	Dementia, mild or unspecified	N/A
HCC135	Drug use with psychotic complications	HCC136, HCC137, HCC138, HCC139
HCC136	Alcohol use with psychotic complications	HCC137, HCC138, HCC139
HCC137	Drug use disorder, moderate/severe, or drug use with non-psychotic complications	HCC138, HCC139
HCC138	Drug use disorder, mild, uncomplicated, except cannabis	HCC139
HCC139	Alcohol use disorder, moderate/severe, or alcohol use with specified non-psychotic complications	N/A
HCC151	Schizophrenia	HCC152, HCC153, HCC154, HCC155
HCC152	Psychosis, except schizophrenia	HCC153, HCC154, HCC155
HCC153	Personality disorders; anorexia/bulimia nervosa	HCC154, HCC155
HCC154	Bipolar disorders without psychosis	HCC155
HCC155	Major depression, moderate or severe, without psychosis	N/A
HCC180	Quadriplegia	HCC181, HCC182, HCC253, HCC254
HCC181	Paraplegia	HCC182, HCC254
HCC182	Spinal cord disorders/injuries	N/A

HCC	HCC Description	Competing Events
HCC190	Amyotrophic lateral sclerosis and other motor neuron disease, spinal muscular atrophy	N/A
HCC191	Quadriplegic cerebral palsy	HCC180, HCC181, HCC182, HCC192, HCC253, HCC254
HCC192	Cerebral palsy, except quadriplegic	HCC180, HCC181, HCC182, HCC253, HCC254
HCC193	Chronic inflammatory demyelinating polyneuropathy and multifocal motor neuropathy	N/A
HCC195	Myasthenia gravis with (acute) exacerbation	HCC196
HCC196	Myasthenia gravis without (acute) exacerbation and other myoneural disorders	N/A
HCC197	Muscular dystrophy	N/A
HCC198	Multiple sclerosis	N/A
HCC199	Parkinson and other degenerative disease of basal ganglia	N/A
HCC200	Friedreich and other hereditary ataxias; Huntington disease	N/A
HCC201	Seizure disorders and convulsions	N/A
HCC202	Coma, brain compression/anoxic damage	N/A
HCC211	Respirator dependence/tracheostomy status/complications	HCC212, HCC213
HCC212	Respiratory arrest	HCC213
HCC213	Cardio-respiratory failure and shock	N/A
HCC221	Heart transplant status/complications	HCC222, HCC223, HCC224, HCC225, HCC226, HCC227
HCC222	End stage heart failure	HCC223, HCC224, HCC225, HCC226, HCC227
HCC223	Heart failure with heart assist device/artificial heart	HCC224, HCC225, HCC226, HCC227
HCC224	Acute on chronic heart failure	HCC225, HCC226, HCC227
HCC225	Acute heart failure (excludes acute on chronic)	HCC226, HCC227

HCC	HCC Description	Competing Events
HCC226	Heart failure, except end stage and acute	HCC227
HCC227	Cardiomyopathy/myocarditis	N/A
HCC228	Acute myocardial infarction	HCC229
HCC229	Unstable angina and other acute ischemic heart disease	N/A
HCC238	Specified heart arrhythmias	N/A
HCC248	Intracranial hemorrhage	HCC249
HCC249	Ischemic or unspecified stroke	N/A
HCC253	Hemiplegia/hemiparesis	HCC254
HCC254	Monoplegia, other paralytic syndromes	N/A
HCC263	Atherosclerosis of arteries of the extremities with ulceration or gangrene	HCC264, HCC383, HCC409
HCC264	Vascular disease with complications	N/A
HCC267	Deep vein thrombosis and pulmonary embolism	N/A
HCC276	Lung transplant status/complications	HCC277, HCC278, HCC279, HCC280
HCC277	Cystic fibrosis	HCC278, HCC279, HCC280
HCC278	Idiopathic pulmonary fibrosis and lung involvement in systemic sclerosis	HCC279, HCC280
HCC279	Severe persistent asthma	HCC280
HCC280	Chronic obstructive pulmonary disease, interstitial lung disorders, and other chronic lung disorders	N/A
HCC282	Aspiration and specified bacterial pneumonias	HCC283
HCC283	Empyema, lung abscess	N/A
HCC298	Severe diabetic eye disease, retinal vein occlusion, and vitreous hemorrhage	N/A
HCC300	Exudative macular degeneration	N/A
HCC326	Chronic kidney disease, stage 5	HCC327, HCC328, HCC329
HCC327	Chronic kidney disease, severe (stage 4)	HCC328, HCC329

HCC	HCC Description	Competing Events
HCC328	Chronic kidney disease, moderate (stage 3B)	HCC329
HCC329	Chronic kidney disease, moderate (stage 3, except 3B)	N/A
HCC379	Pressure ulcer of skin with necrosis through to muscle, tendon, or bone	HCC380, HCC381, HCC382, HCC383
HCC380	Chronic ulcer of skin, except pressure, through to bone or muscle	HCC381, HCC382, HCC383
HCC381	Pressure ulcer of skin with full thickness skin loss	HCC382, HCC383
HCC382	Pressure ulcer of skin with partial thickness skin loss	HCC383
HCC383	Chronic ulcer of skin, except pressure, not specified as through to bone or muscle	N/A
HCC385	Severe skin burn	N/A
HCC387	Pemphigus, pemphigoid, and other specified autoimmune skin disorders	N/A
HCC397	Major head injury with loss of consciousness (> 1 hour)	HCC202, HCC398, HCC399
HCC398	Major head injury with loss of consciousness (< 1 hour or unspecified)	HCC202, HCC399
HCC399	Major head injury without loss of consciousness	N/A
HCC401	Vertebral fractures without spinal cord injury	N/A
HCC402	Hip fracture/dislocation	N/A
HCC405	Traumatic amputations and complications	HCC409
HCC409	Amputation status, lower limb/amputation complications	N/A
HCC454	Stem cell, including bone marrow, transplant status/complications	N/A
HCC463	Artificial openings for feeding or elimination	N/A

6. Additional simulation results

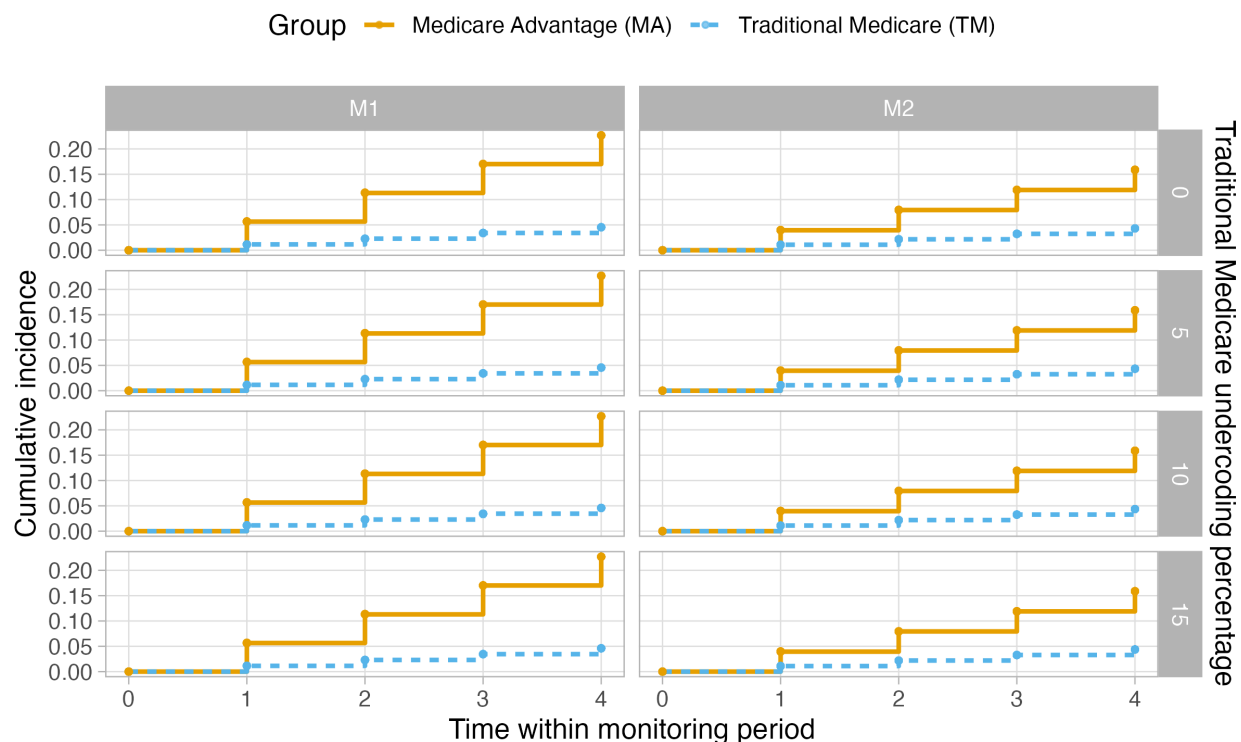


Figure 1: **Cumulative incidence functions for the Specified Heart Arrhythmias Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 25% any-available MA upcoding.** Specified heart arrhythmias corresponds to HCC238, which does not have any competing events. For this HCC, 25% of any-available individuals in the MA group are upcoded and 5% of any-available individuals in the TM comparison group are upcoded. The first monitoring period is labeled M1, and the second monitoring period is labeled M2. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

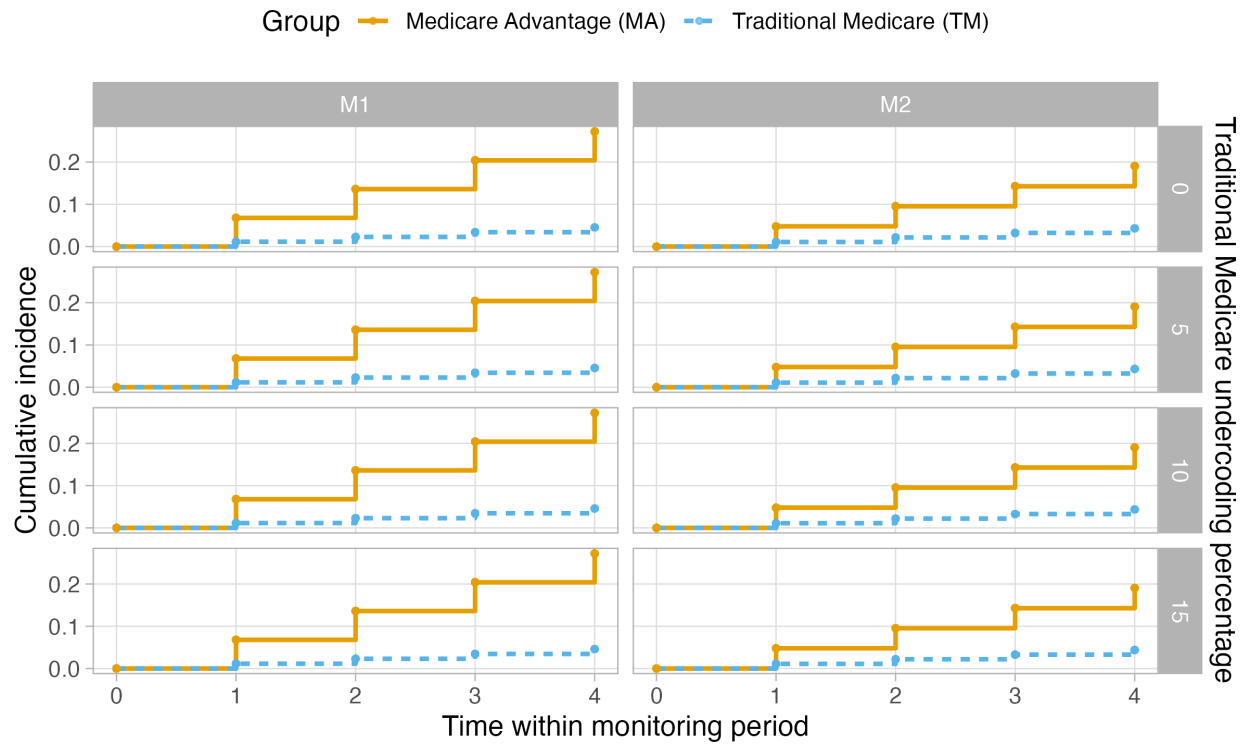


Figure 2: **Cumulative incidence functions for the Specified Heart Arrhythmias Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 30% any-available MA upcoding.** Specified heart arrhythmias corresponds to HCC238, which does not have any competing events. For this HCC, 30% of any-available individuals in the MA group are upcoded and 5% of any-available individuals in the TM comparison group are upcoded. The first monitoring period is labeled M1, and the second monitoring period is labeled M2. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

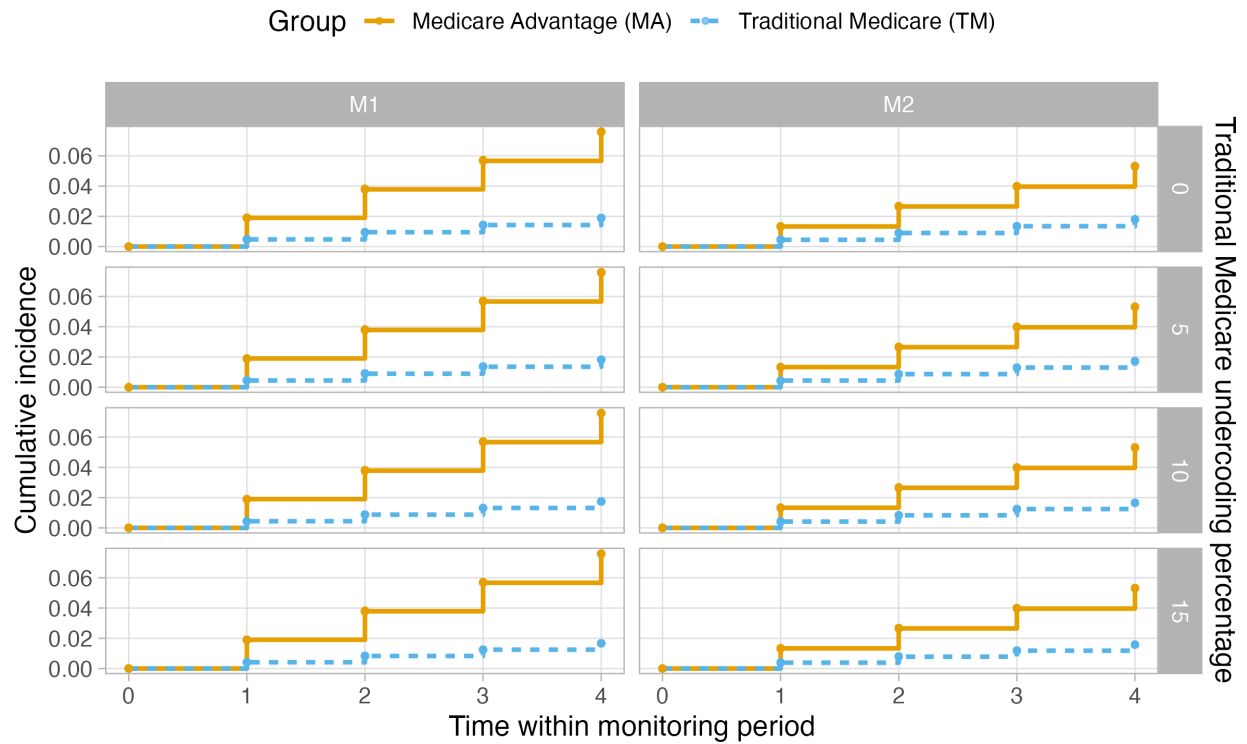


Figure 3: **Cumulative incidence functions for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 20% lower severity MA upcoding.** ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). 20% of any individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded and 5% of individuals in the TM comparison group are upcoded similarly. The first monitoring period is labeled M1, and the second monitoring period is labeled M2. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

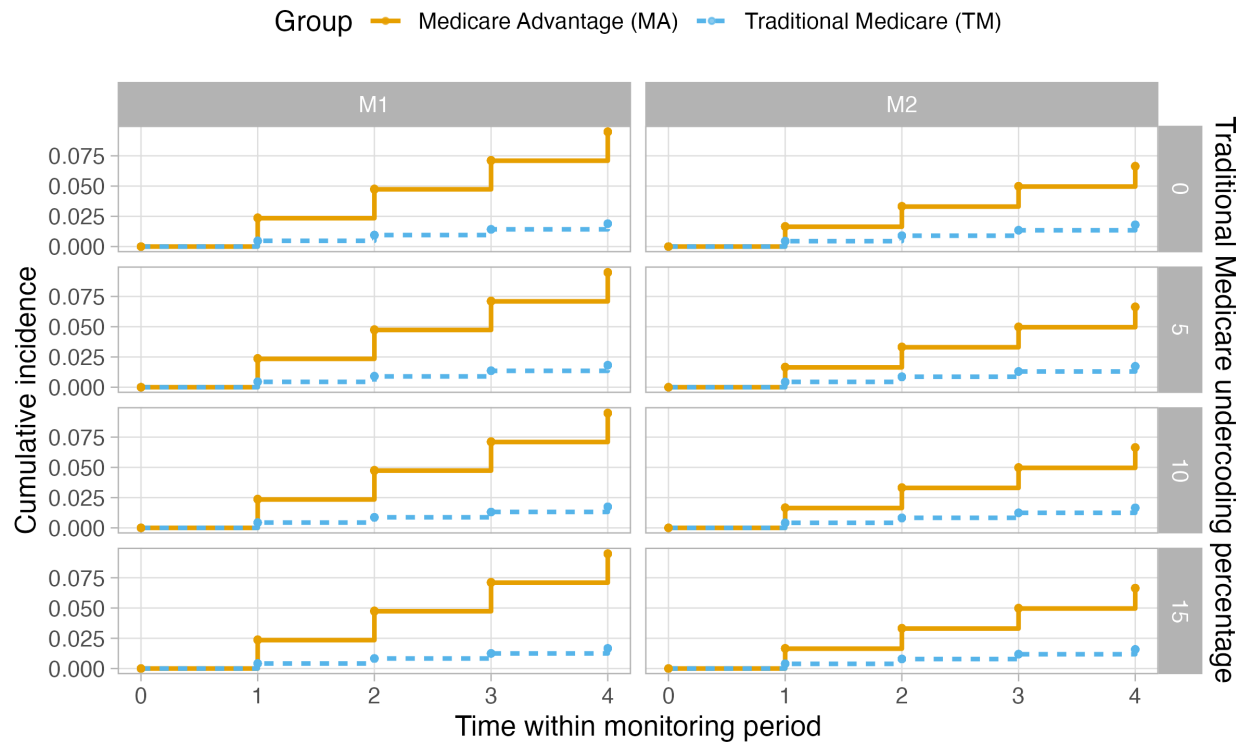


Figure 4: **Cumulative incidence functions for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 25% lower severity MA upcoding.** ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). 25% of any individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded and 5% of individuals in the TM comparison group are upcoded similarly. The first monitoring period is labeled M1, and the second monitoring period is labeled M2. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

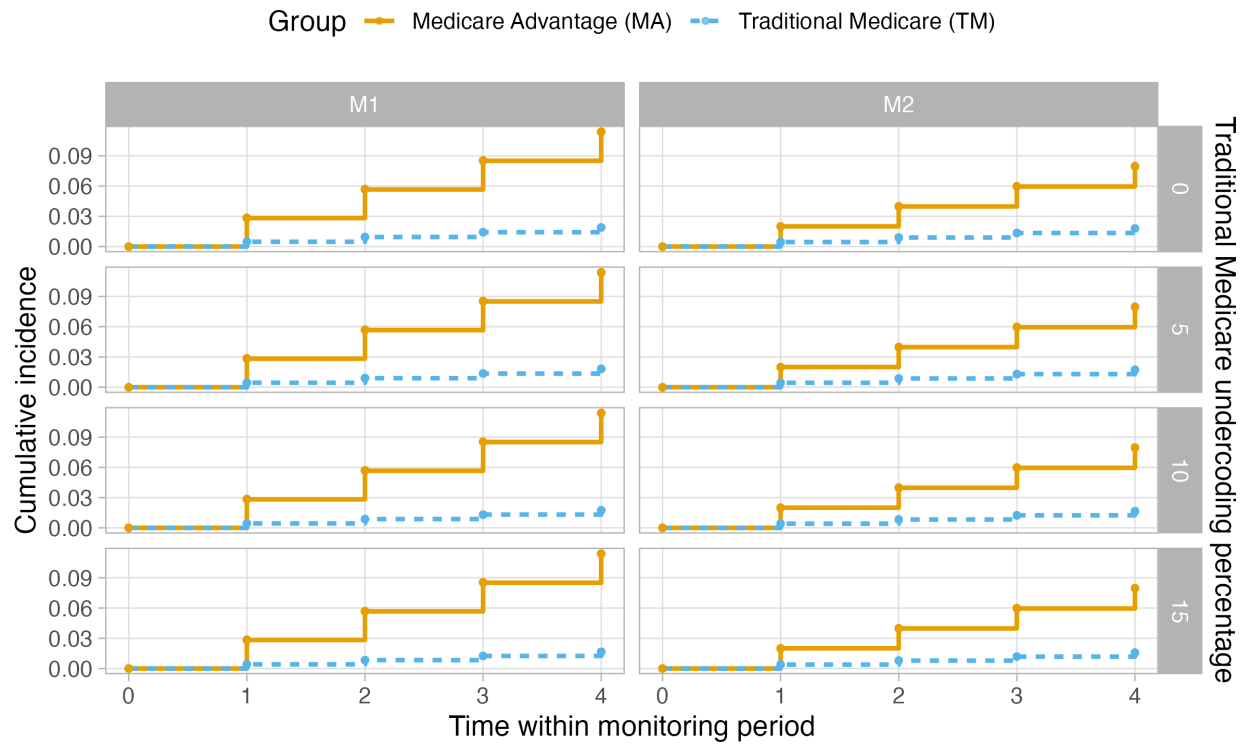


Figure 5: **Cumulative incidence functions for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 30% lower severity MA upcoding.** ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). 30% of any individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded and 5% of individuals in the TM comparison group are upcoded similarly. The first monitoring period is labeled M1, and the second monitoring period is labeled M2. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

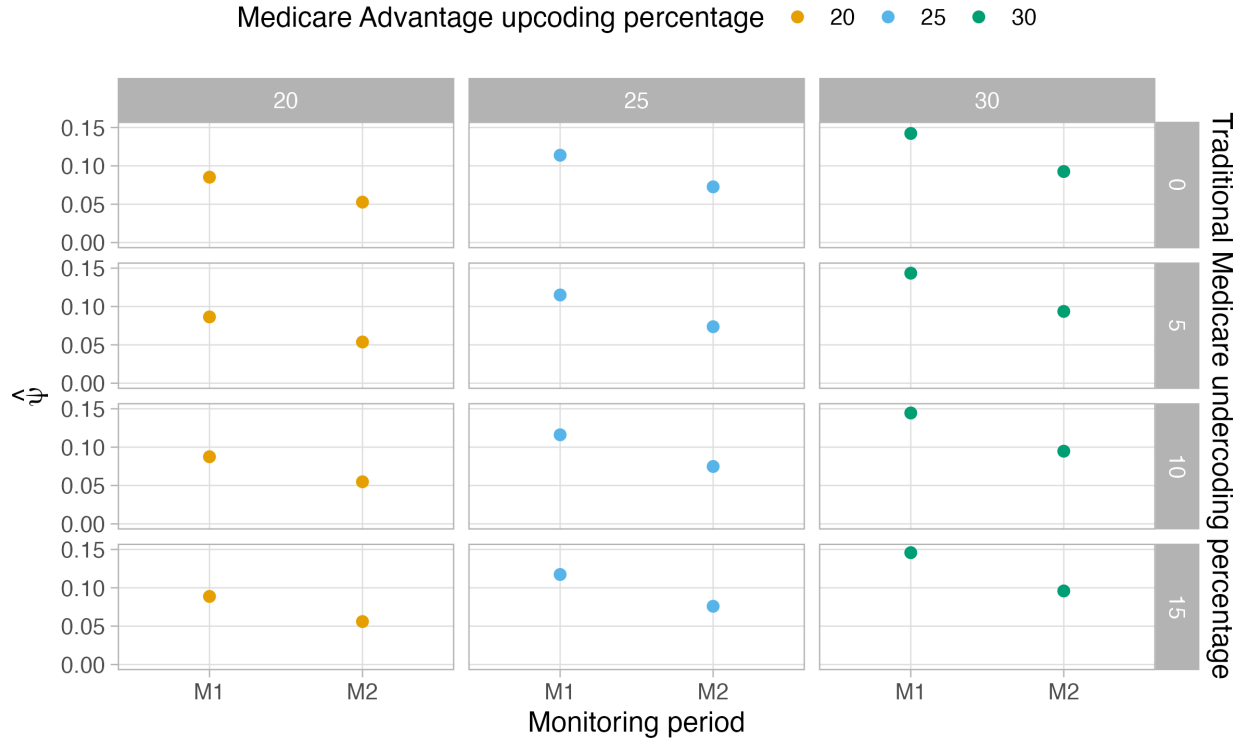


Figure 6: **Within-monitoring period ψ estimates for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups.** $\hat{\psi}$ corresponds to the proposed estimator for ψ , or the difference in mean time without event (i.e. incident coding of the HCC) across groups. ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). For this HCC, individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded to varying degrees and individuals in the TM comparison group are up-coded 5% similarly. The first monitoring period is labeled M1, and the second monitoring period is labeled M2. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

References

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